

Measurement

Marking Scheme & Blueprint

MTH1W · Module 7 Test — Teacher Copy

Blueprint

Curriculum: MTH1W (Grade 9 de-streamed math). Total: 70 marks · 60 minutes · 30 questions. Categories: K/U, Application (App), Thinking (Think), Communication (Comm). $\pi \approx 3.14$; π -values rounded to 1 d.p.

Q	Sub-topic	Level	Marks	Category	Section
A1	Rectangle Area	L1	1	K/U	A
A2	Rectangle Perimeter	L1	1	K/U	A
A3	Square Area	L1	1	K/U	A
A4	Triangle Area	L1	1	K/U	A
A5	Cube Volume	L1	1	K/U	A
A6	Cylinder Formula	L1	1	K/U	A
A7	Circumference	L1	1	K/U	A
A8	Circle Formula	L1	1	K/U	A
A9	Square Perimeter	L1	1	K/U	A
A10	Parallelogram Area	L1	2	K/U	A
A11	Trapezoid Area	L1	2	K/U	A
A12	Circle Area	L1	2	K/U	A
A13	Prism Volume	L1	2	K/U	A
A14	Cube Surface Area	L1	2	K/U	A
A15	Cylinder Volume	L1	2	K/U	A
B16	Triangle Area	L2	2	App	B
B17	Prism Surface Area	L2	2	App	B
B18	Cylinder Volume	L2	2	App	B
B19	Cone Volume	L2	2	App	B
B20	Sphere Surface Area	L2	2	App	B
B21	Pyramid Volume	L2	2	App	B
B22	Cylinder Surface Area	L2	3	App	B
B23	Composite Area	L2	3	App	B
B24	Sphere Volume	L2	3	App	B
C25	Capacity	L3	4	Think	C
C26	Composite Solid	L3	4	Think	C
C27	Recasting Solids	L3	5	Think	C
C28	Slant & Surface Area	L3	5	Think	C
C29	Silo (Cylinder + Cone)	L3	6	Think	C
D30	Optimization Reasoning	L3	4	Comm	D

By category (marks): K/U 21 · Application 21 · Thinking 24 · Communication 4

By level (marks): L1 21 · L2 21 · L3 28 → L1 30% · L2 30% · L3 40%

Worked Solutions & Mark Allocation

- 1. (A1 · K/U · 1 marks)**
 $A = 7 \times 6 = 42 \text{ cm}^2$. (A1)
- 2. (A2 · K/U · 1 marks)**
 $P = 2(8+3) = 22 \text{ cm}$. (A1)
- 3. (A3 · K/U · 1 marks)**
 $A = 9^2 = 81 \text{ cm}^2$. (A1)
- 4. (A4 · K/U · 1 marks)**
 $A = \frac{1}{2} \times 8 \times 5 = 20 \text{ cm}^2$. (A1)
- 5. (A5 · K/U · 1 marks)**
 $V = 3^3 = 27 \text{ cm}^3$. (A1)
- 6. (A6 · K/U · 1 marks)**
 $V = \pi r^2 h$. (A1)
- 7. (A7 · K/U · 1 marks)**
 $C = 2 \times 3.14 \times 10 = 62.8 \text{ cm}$. (A1)
- 8. (A8 · K/U · 1 marks)**
 $A = \pi r^2$. (A1)
- 9. (A9 · K/U · 1 marks)**
 $P = 4 \times 12 = 48 \text{ cm}$. (A1)
- 10. (A10 · K/U · 2 marks)**
 $A = \text{base} \times \text{height} = 11 \times 4 = 44 \text{ cm}^2$. (M1, A1)
- 11. (A11 · K/U · 2 marks)**
 $A = \frac{1}{2}(8+4)(6) = 36 \text{ cm}^2$. (M1, A1)
- 12. (A12 · K/U · 2 marks)**
 $A = 3.14 \times 4^2 = 50.2 \text{ cm}^2$. (M1, A1)
- 13. (A13 · K/U · 2 marks)**
 $V = 5 \times 4 \times 6 = 120 \text{ cm}^3$. (M1, A1)
- 14. (A14 · K/U · 2 marks)**
 $SA = 6 \times 7^2 = 294 \text{ cm}^2$. (M1, A1)
- 15. (A15 · K/U · 2 marks)**
 $V = 3.14 \times 3^2 \times 5 = 141.3 \text{ cm}^3$. (M1, A1)
- 16. (B16 · Application · 2 marks)**
 $A = \frac{1}{2} \times 14 \times 9 = 63 \text{ cm}^2$. (M1, A1)

17. (B17 • Application • 2 marks)

$$SA = 2(8 \times 5 + 8 \times 4 + 5 \times 4) = 2(40 + 32 + 20) = 184 \text{ cm}^2. \text{ (M1, A1)}$$

18. (B18 • Application • 2 marks)

$$V = 3.14 \times 6^2 \times 10 = 1130.4 \text{ cm}^3. \text{ (M1, A1)}$$

19. (B19 • Application • 2 marks)

$$V = (1/3)(3.14)(3^2)(8) = 75.4 \text{ cm}^3. \text{ (M1, A1)}$$

20. (B20 • Application • 2 marks)

$$SA = 4 \times 3.14 \times 7^2 = 615.4 \text{ cm}^2. \text{ (M1, A1)}$$

21. (B21 • Application • 2 marks)

$$V = (1/3) \times 5^2 \times 12 = (1/3) \times 25 \times 12 = 100 \text{ cm}^3. \text{ (M1, A1)}$$

22. (B22 • Application • 3 marks)

$$\begin{aligned} r = 4: SA &= 2\pi r^2 + 2\pi rh = 2 \times 3.14 \times 4^2 + 2 \times 3.14 \times 4 \times 9. \text{ (M1, M1)} \\ &= 100.48 + 226.08 = 326.6 \text{ cm}^2. \text{ (A1)} \end{aligned}$$

23. (B23 • Application • 3 marks)

$$\begin{aligned} \text{Rectangle } 14 \times 8 &= 112; \text{ semicircle } r = 4: \frac{1}{2} \times 3.14 \times 4^2 = 25.12. \text{ (M1, M1)} \\ \text{Total} &= 112 + 25.12 = 137.1 \text{ cm}^2. \text{ (A1)} \end{aligned}$$

24. (B24 • Application • 3 marks)

$$\begin{aligned} V &= (4/3)\pi r^3 = (4/3)(3.14)(5^3) = (4/3)(3.14)(125). \text{ (M1, M1)} \\ &= 523.3 \text{ cm}^3. \text{ (A1)} \end{aligned}$$

25. (C25 • Thinking • 4 marks)

$$\begin{aligned} V &= 3.14 \times 2^2 \times 3 = 37.7 \text{ m}^3. \text{ (M1, A1)} \\ \text{Litres} &= 37.68 \times 1000 = 37680 \text{ L. (M1, A1)} \end{aligned}$$

26. (C26 • Thinking • 4 marks)

$$\begin{aligned} \text{Cube: } 5^3 &= 125; \text{ pyramid: } (1/3) \times 5^2 \times 9 = 75. \text{ (M1, M1)} \\ \text{Total} &= 125 + 75 = 200 \text{ cm}^3. \text{ (A1, A1)} \end{aligned}$$

27. (C27 • Thinking • 5 marks)

$$\begin{aligned} \text{Cylinder } V &= 3.14 \times 5^2 \times 20 = 1570.0 \text{ cm}^3. \text{ (M1, A1)} \\ \text{Sphere } V &= (4/3)(3.14)(5^3) = 523.3 \text{ cm}^3. \text{ (M1)} \\ \text{Number} &= 1570 \div 523.3 = 3 \text{ complete spheres. (M1, A1)} \end{aligned}$$

28. (C28 • Thinking • 5 marks)

$$\begin{aligned} \text{Slant } l &= \sqrt{9^2 + 12^2} = \sqrt{225} = 15 \text{ cm. (M1, A1)} \\ SA &= \pi r^2 + \pi rl = 3.14 \times 9^2 + 3.14 \times 9 \times 15 = 254.34 + 423.9. \text{ (M1)} \\ &= 678.2 \text{ cm}^2. \text{ (A1, A1)} \end{aligned}$$

29. (C29 • Thinking • 6 marks)

$$\begin{aligned} \text{Cylinder } V &= 3.14 \times 4^2 \times 9 = 452.2 \text{ m}^3. \text{ (M1, A1)} \\ \text{Cone } V &= (1/3)(3.14)(4^2)(3) = 50.2 \text{ m}^3. \text{ (M1, A1)} \\ \text{Total} &= 452.16 + 50.24 = 502.4 \text{ m}^3. \text{ (M1, A1)} \end{aligned}$$

30. (D30 · Communication · 4 marks)

The sphere has the smaller surface area. (A1)

For a fixed volume, the sphere is the solid with the least surface area of all; among boxes the cube is best, but the sphere beats the cube. (C1)

General principle: for a given volume the surface area is minimized by the most 'compact' shape — the sphere. (C1)

This is why bubbles and water droplets form spheres. (C1)